Noor

See, Speak, and Stay Safe—instantly

Mission Statement: Real-time assistance application for people with visual and hearing impairments using a reliable display.

Team: PyBlaze

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The Issue We're Solving

Navigation Barriers

People with visual impairments struggle with real-time obstacle detection, reading signs, and understanding their surroundings safely. The rate of visual impairment reached 1.36% in 2022.

Communication Gaps

Deaf individuals face barriers converting between sign language, speech, and text in real-time conversations. The deaf and mute constitute about 12% of the total population in Jordan.

Safety Concerns

Caregivers worry about their loved ones' safety but lack real-time awareness without compromising independence.

Three Connected Experiences

Blind Mode

Real-time scene description, OCR text reading powered, and voice guidance.

Deaf Mode

Bidirectional sign language conversion with high-contrast UI, vibration alerts, and seamless text-to-voice communication.

Monitor Dashboard

Secure caregiver access with location sharing and private communication channels.

Blind Mode: Your Digital Eyes

Real-Time Scene Understanding

It provides real-time scene descriptions and reads signs and other text aloud, with all features controlled by voice commands.

Noor technical flow

Text recognition and reading

- Google Vision API
- Google Text-to-Speech

Object detection with precise

- DeepFace
- Yolo
- Gemini
- Text-to-Speech and Speech-to-Text

Deaf Mode: Breaking Communication Barriers

Speech to Sign

Converts spoken phrases into visual sign language representations.

Sign to Text/Voice

Recognizes sign language and translates it to text or spoken words for others.

Noor technical flow

- Yolo fine tunning on ASL
- Gemini API
- Google cloud storage
- Firebase

A high-contrast interface with large controls and vibration feedback ensures accessibility in all lighting conditions.

Monitor Dashboard

01

Continuous Heartbeat

Background service monitors user status and location every few seconds automatically.

02

Last Known Location

Emergency location sharing helps caregivers respond quickly while preserving daily privacy.

The Future Plan: Smart Glasses

Noor on Smart Glasses

Hands-Free Operation	Continuous Awareness
Camera, bone-conduction audio, and haptic feedback in a wearable form factor.	Real-time scene understanding and heads-up navigation cues without holding a device.

Same Privacy Model

Maintains secure pairing with caregivers and a more natural interface.

This is our primary evolution path—transforming smartphones into seamless, wearable assistance.

Thank you for your times!

Any Questions or Feedback?

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